Question:
There exist real numbers x and y, both greater than 1, such that $\log_x\left(y^x\right)=\log_y\left(x^{4y}\right)=10$. Find xy.

Reasoning + Answer:

We are given that there exist real numbers x and y, both greater than 1, such that:

$$
\log_x(y^x) = 10 \quad \text{ and } \quad \log_y(x^{4y}) = 10
$$

We are to find xy.

Step 1: Simplify the first equation

$$
\log_x(y^x) = 10
$$

Use the logarithmic identity: $\log_b(a^c) = c \log_b a$

So:

$$
\log_x(y^x) = x \log_x y = 10
$$

Therefore:

$$
x \log_x y = 10 \quad \text{(1)}
$$

Step 2: Simplify the second equation

$$
\log_y(x^{4y}) = 10
$$

Again, $\log_y(x^{4y}) = 4y \log_y x = 10$

So:

$$
4y \log_y x = 10 \quad \text{(2)}
$$

Step 3: Use change of base

LetâĢĻs write $\log_x y$ and $\log_y x$ in terms of each other.

We know that:
$$
\log_x y = \frac{1}{\log_y x}
$$

LetâĢĻs denote $a = \log_x y$. Then $\log_y x = \frac{1}{a}$

From equation (1):
$$
x \cdot a = 10 \quad \Rightarrow \quad a = \frac{10}{x} \quad \text{(3)}
$$

From equation (2):
$$
4y \cdot \frac{1}{a} = 10 \quad \Rightarrow \quad \frac{4y}{a} = 10 \quad \Rightarrow \quad \frac{1}{a} = \frac{10}{4y} = \frac{5}{2y} \quad \text{(4)}
$$

Now from (3), $a = \frac{10}{x}$, so $\frac{1}{a} = \frac{x}{10}$

But from (4), $\frac{1}{a} = \frac{5}{2y}$

So equate them:

$$
\frac{x}{10} = \frac{5}{2y}
$$

Multiply both sides by 10:

$$
x = \frac{50}{2y} = \frac{25}{y}
$$

So:

$$
x = \frac{25}{y} \quad \text{(5)}
$$

Now recall from equation (1): $xa = 10$, and $a = \log_x y$

We already have $x = \frac{25}{y}$, so now express a in terms of x and y

From $x = \frac{25}{y}$, we get $y = \frac{25}{x}$

So now we can express $\log_x y = a$, and we know from (1): $xa = 10$

But $y = \frac{25}{x}$, so $a = \log_x y = \log_x \left(\frac{25}{x} \right)$

LetâĢĻs compute that:

$$
\log_x \left(\frac{25}{x} \right) = \log_x 25 - \log_x x = \log_x 25 - 1
$$

So:
$$
a = \log_x 25 - 1
$$

From equation (1): $xa = 10$

So:
$$